

Exit Timing: The Trapped Investor Problem

Session 11 • Unit 3 • Optimal stopping with stochastic L_t

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Today: when should an LP exit a PE position?

1

Optimal stopping framework

Continue holding vs sell now

2

Forced vs discretionary exit

Regulatory • operational • optimization

3

The four trapped-investor conditions

When exit at fair value is impossible

4

Smooth pasting (intuition)

T2 derives in S25 • T1 sees the result here

5

LAV operator: value of holding

Compute the continuation value

The LP's exit decision as optimal stopping

At each time t , the LP faces a binary choice:

Option A: EXIT NOW

Receive $P^{\text{secondary}}(L_t)$

The secondary market clearing price at current liquidity state. Discount to NAV reflects current L_t regime.

Option B: CONTINUE HOLDING

Receive $E[V^{\text{LAV}} | L_t]$

Expected future value under LAV operator. Better expected outcome if L_t is currently depressed.

Exit when:

$$P^{\text{secondary}}(L_t) > E[V^{\text{LAV}} | L_t]$$

This defines the exit boundary $L^(t)$ we'll plot next session.*

Three types of LP exit decisions

Forced — regulatory

Mandatory rebalancing

Solvency II capital trigger.
ERISA allocation cap breach.
Public pension cash-out duty.

LP has no choice. Sells at any discount.

→ Common in 2008-09

Forced — operational

Cash flow distress

Endowment spending need.
Insurance liability claim.
Unexpected cash shortfall.

LP must liquidate at market.

→ Hits Yale, Harvard 2009

Discretionary — optimization

Tactical exit

GE-LAV-optimal exit decision.
Based on L_t state + horizon.
Can wait for better regime.

The focus of this session.

→ GE-LAV value-add

GE-LAV's exit boundary $L^*(t)$ is the optimal discretionary rule — assuming you have the option to wait.

Four conditions that trap an LP

Condition 1

Forced sale during low-L regime

Regulatory trigger forces sale exactly when L_t is depressed. LP sells at deep discount with no flexibility.

Condition 2

GP gates redemptions

GP imposes redemption gates or extends fund life. LP cannot exit even if they wish to.

Condition 3

Secondary market freezes

Crisis depths see secondary bid disappear (2008 Q4). No price discovery. LP is paper-illiquid.

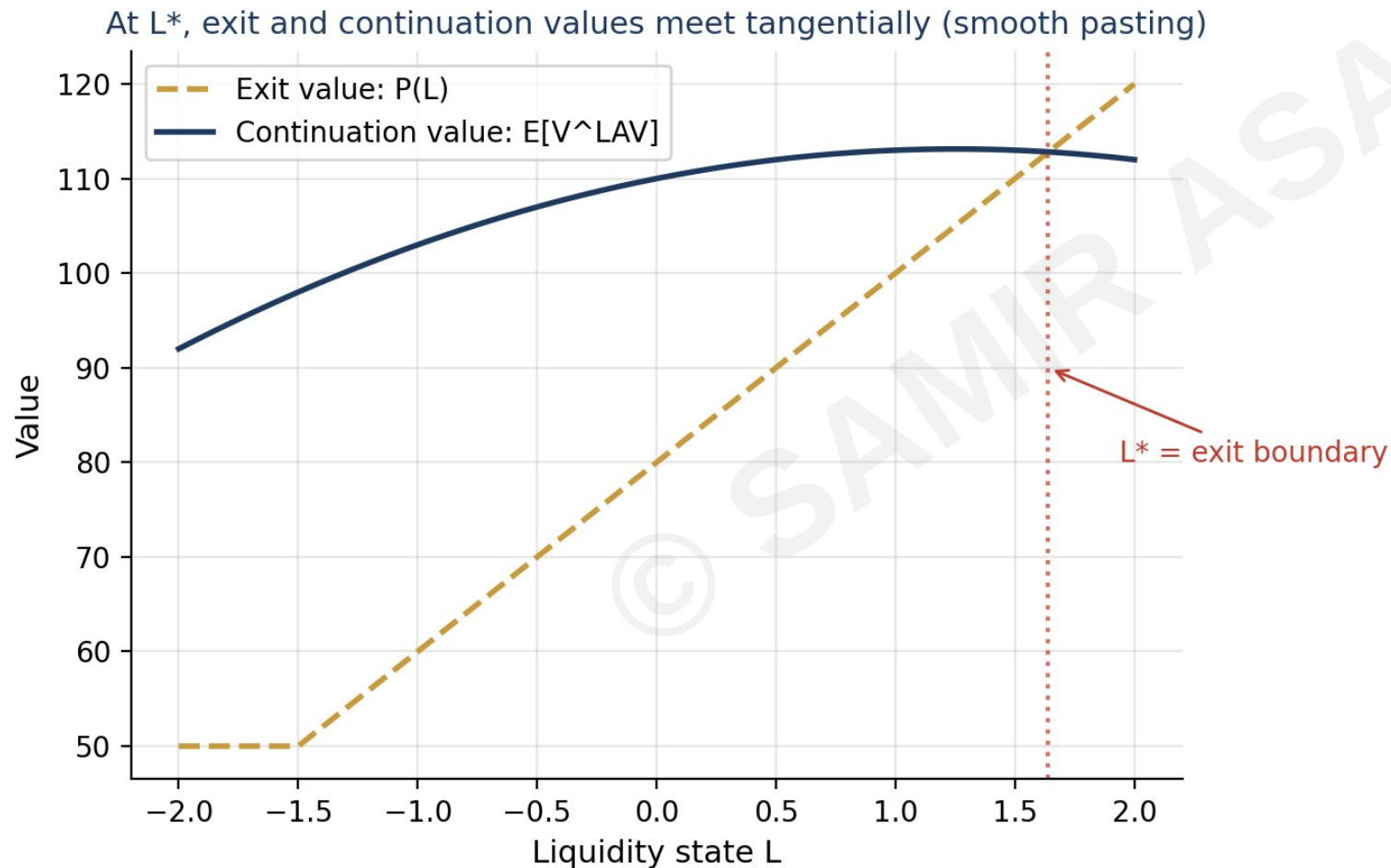
Condition 4

Correlation traps

LP's other assets force allocation rebalancing. Public markets crash → private allocation looks too large → forced sale.

GE-LAV cannot remove trap conditions — but it can predict when they will bind and pre-emptively reduce exposure.

The smooth pasting condition (intuition)



Smooth pasting

- ▶ At L^* : exit and hold values are equal AND have equal slopes.
- ▶ Above L^* : exit value dominates → SELL.
- ▶ Below L^* : continuation value dominates → HOLD.
- ▶ Track 2 derives this via free-boundary HJB equation in Session 25.
- ▶ Track 1 uses the boundary as given.

Computing the value of holding: a worked example

Setup: PE position, current $L = -0.5$ (mild stress), 3 years remaining

Step	Action	Result
1	Simulate 10,000 OU paths starting from $L_0 = -0.5$	Distribution of L_3
2	Compute $r(L_t) = 0.045 - 0.025L + 0.021L^2$ along each path	Path-dependent rate
3	Discount cash flows along each path: $e^{(-\int r dt)}$	Path PV
4	Average across paths to get $E[V^{LAV} L=-0.5]$	Continuation value
5	Compare to secondary market price at $L=-0.5$	Hold vs exit decision

Numerical result for this setup

$P^{\text{secondary}}$ at $L=-0.5$: $\approx \$82\text{M}$ $E[V^{LAV} | L=-0.5]$: $\approx \$94\text{M}$ Decision: HOLD

Session 11 wrap-up

Key takeaways

1

LP exit is an optimal stopping problem: $P^{\text{secondary}}$ vs $E[V^{\text{LAV}}]$.

2

Three exit types: forced regulatory, forced operational, discretionary.

3

Four conditions that trap LPs: regulation, gates, freezes, correlations.

4

Smooth pasting: at L^* , exit and continuation values are tangent.

5

GE-LAV computes the continuation value via path-dependent simulation.

NEXT: SESSION 12

Session 12: Explicit calculation and visualization of $L^*(t)$.

We'll generate the exit boundary curve, see how it behaves near maturity, and apply it to historical episodes (GFC 2009, COVID 2020, 2022 rate shock).

Reading: Book Ch 6 §6.5-6.7